

COMP9334

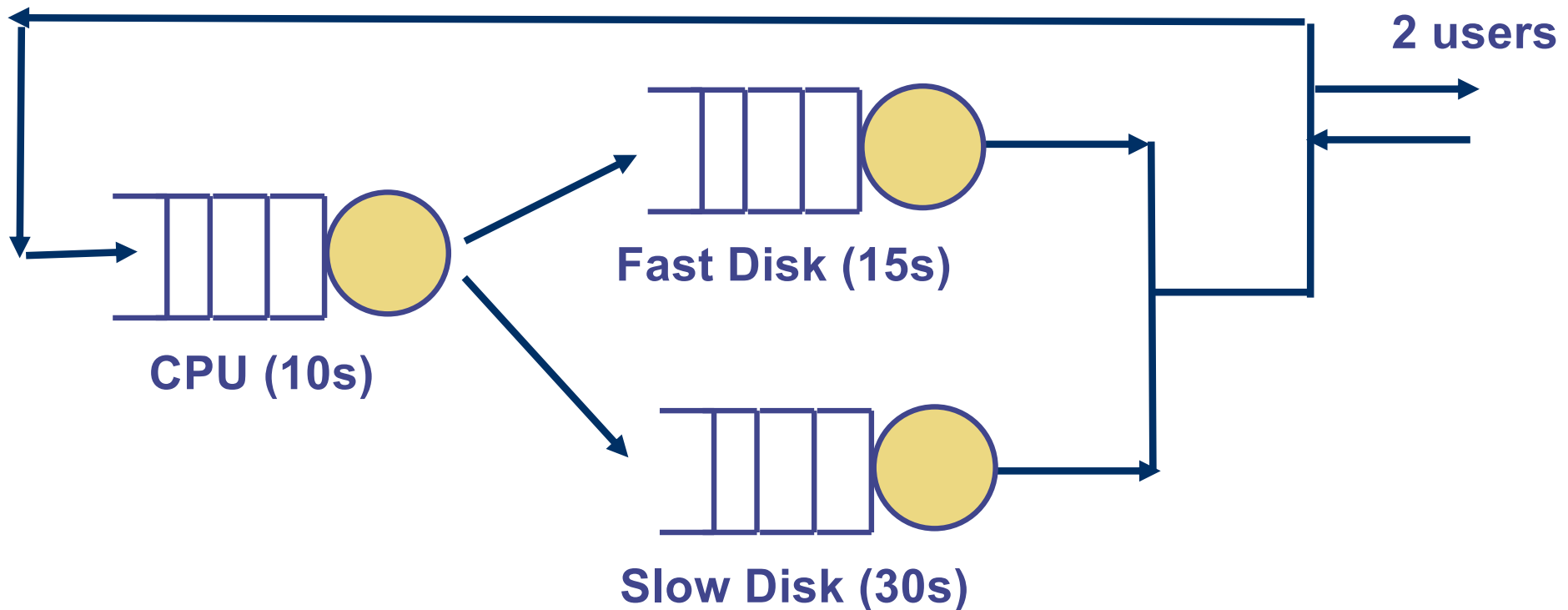
Capacity Planning for Computer Systems and Networks

Week 7B: Mean Value Analysis

This lecture

- Methods to *efficiently* analyse a closed queueing network
- Motivation
 - You have learnt how to analyse a closed queueing network in Week 3B using Markov chain
 - However, the method can only be used for a small number of users
- This week we will study a method that can be used for a larger number of users
- Let us begin by revisiting the database server example in Week 3B

DB server example

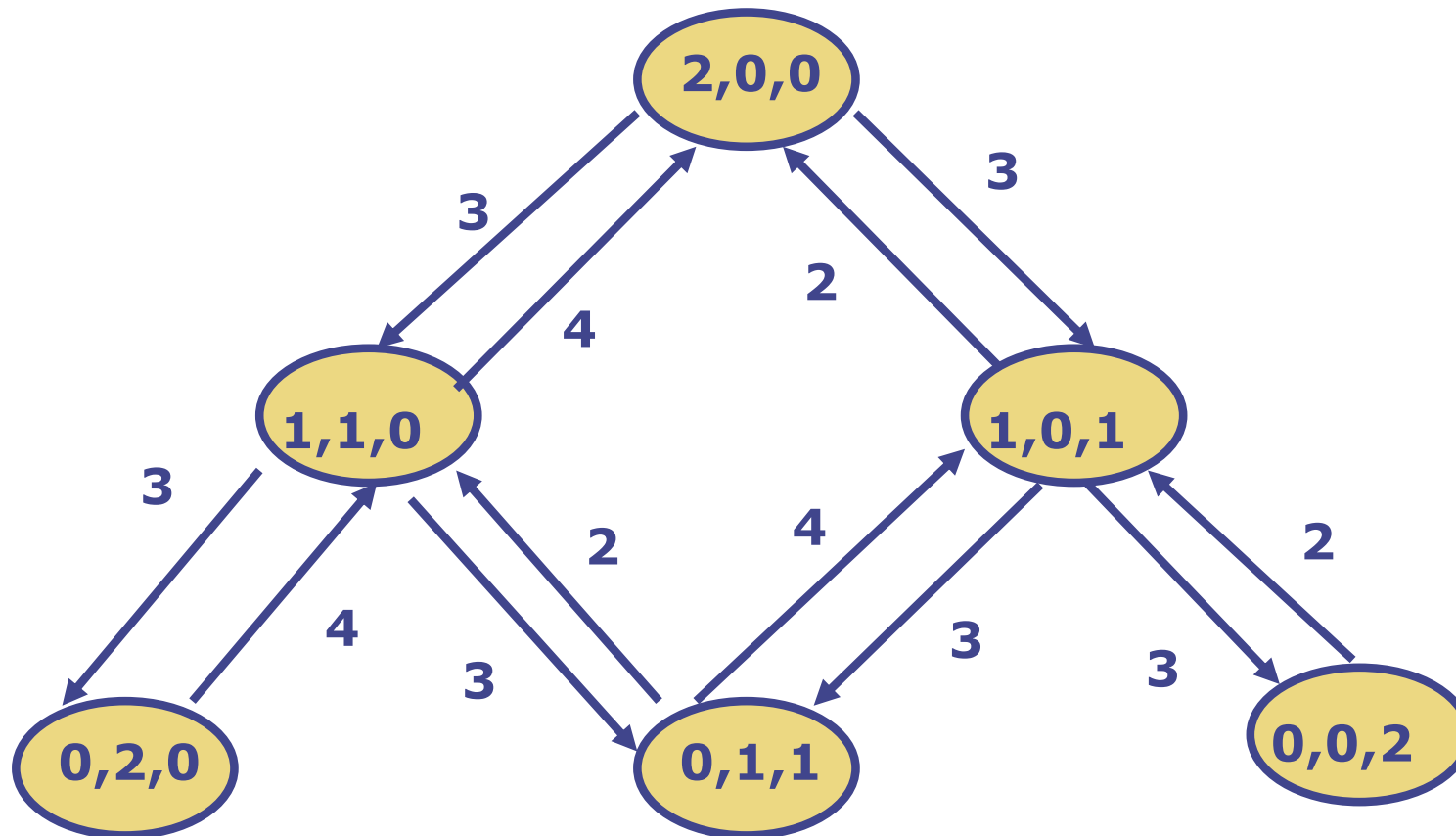


- 1 CPU, 1 fast disk, 1 slow disk.
- Peak demand = 2 users in the system all the time.
- Transactions alternate between CPU and disks.
- The transactions will equally likely find files on either disk
- Service time are exponentially distributed with mean showed in parentheses.

Markov chain solution to the DB server problem

- In Week 3B, we used Markov chain to solve this problem
- We use a 3-tuple (X,Y,Z) as the state
 - X is # users at CPU
 - Y is # users at fast disk
 - Z is # users at slow disk
- Examples
 - $(2,0,0)$: both users at CPU
 - $(1,0,1)$: one user at CPU and one user at slow disk
- Six possible states
 - $(2,0,0)$ $(1,1,0)$ $(1,0,1)$ $(0,2,0)$ $(0,1,1)$ $(0,0,2)$

Markov model for the database server with 2 users



Solving the model

- Solve for the probability in each state $P(2,0,0)$, $P(1,1,0)$, etc.
 - There are 6 states so we need 6 equations
- After solving for $P(2,0,0)$, $P(1,1,0)$ etc. we can find
 - Utilisation
 - Throughput,
 - Response time,
 - Average number of users in each component etc.

What if we have 3 users instead?

- What if we have 3 users in the database example instead of only 2 users?
- We continue to use (X,Y,Z) as the state
 - X is the # users at CPU
 - Y is the # users at the fast disk
 - Z is the # users at the slow disk
- How many states will you need?
- We need 10 states:
 - $(3,0,0)$,
 - $(2,1,0), (2,0,1)$
 - $(1,2,0), (1,1,1), (1,0,2)$
 - $(0,3,0), (0,2,1), (0,1,2), (0,0,3)$

What if there are n users?

- You can show that if there are n users in the database server, the number of states m required will be

$$\frac{(n+1)(n+2)}{2}$$

- For $n = 100$, m (= #states) ~ 50000
- You can automate the computational process but where is the computational bottleneck?
 - Solving a system of m linear equations in m unknowns has a complexity of $O(m^3)$
- For our database server with n users, the computational complexity is about $O(n^6)$

Weaknesses of Markov model

- The Markov model for a practical system will require many states due to
 - Large number of users
 - Large number of components
- Large # states
 - More transitions to identify
 - Though this can be automated
 - If you've m states, you need to solve a set of m equations. A larger set of equation to solve.
 - The complexity of solving a set of m linear equations in m unknowns is $O(m^3)$

Mean value analysis (MVA)

- An iterative method to find the
 - Utilisation
 - Mean throughput
 - Mean response time
 - Mean number of users
- The complexity is approximately $O(nk)$ where
 - n is the number of users
 - k is the number of devices
- The complexity of MVA makes it a very practical method

MVA - overview

- MVA analysis has been derived for
 - Closed model
 - Single-class
 - Multi-class
 - Open model
 - Mixed model with both open and closed queueing
- This lecture discusses MVA for single-class closed model

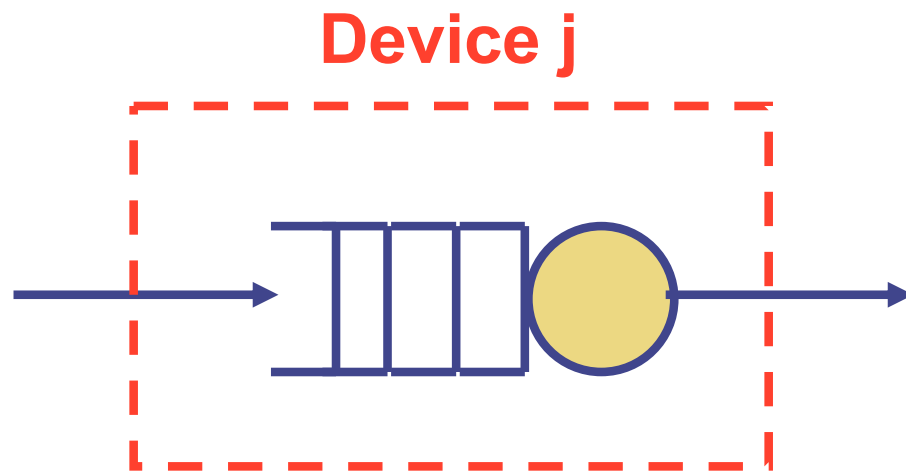
MVA for closed system

- Consider a closed queueing network with a single-class of customers
- You are given a system with K devices
- You are given that each customer
 - Visits device j on average $V(j)$ times
 - Requires a mean service time of $S(j)$ from device j
 - *Note: The service time required is assumed to be exponentially distributed*
- From the information given, we can deduce that the service demand $D(j)$ for device j is $V(j) S(j)$
- How do we obtain $D(j)$ for a practical system?

Key idea behind MVA

- Key idea behind MVA is *iteration*
 - If you know the solution to the problem when there are n customers in the system, you can find the solution when there are $(n+1)$ customers

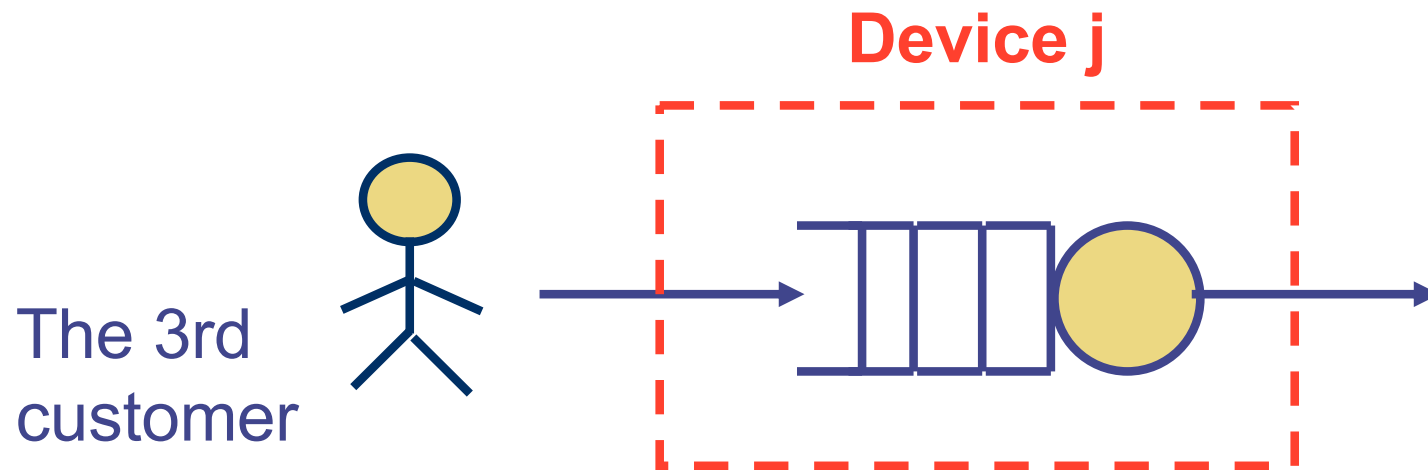
Let us consider a simple example to motivate the iteration in MVA. Consider device j (say) of a queueing network.



Assume that we know when there are 2 customers in the system, the average number of customers in device j is 0.6 (say).

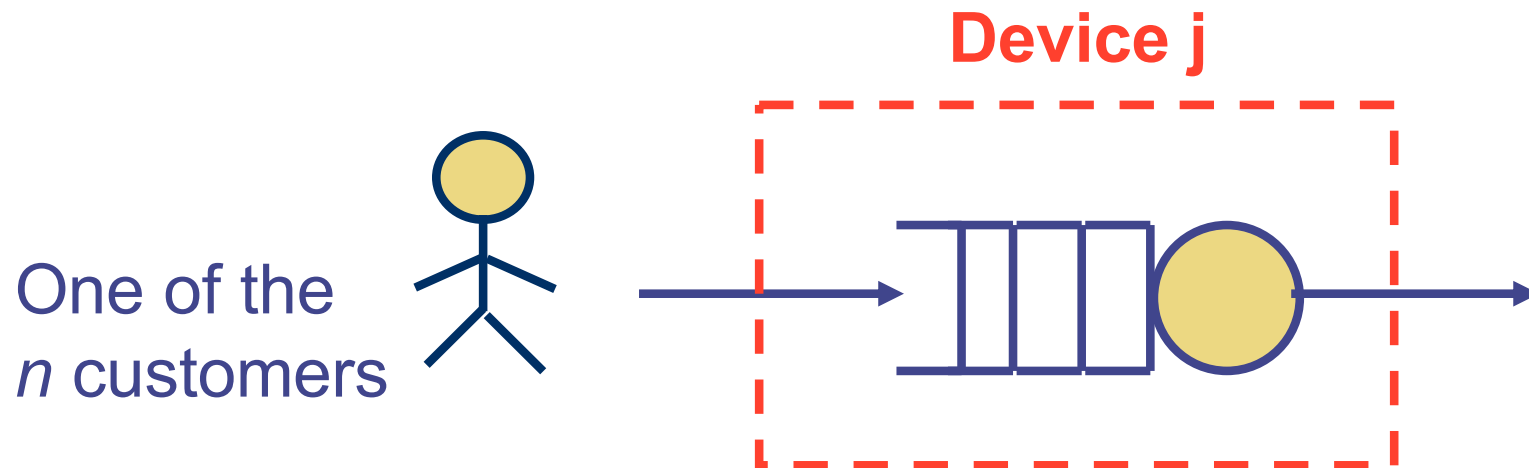
What happens when there are 3 customers?

What happens when there are 3 customers?



- Let us assume the 3rd customer is arriving at device j .
- Where will the other 2 customers be? We cannot tell exactly but we know that there is on average of 0.6 customers in device j when there are 2 customers.
- The 3rd customer will see on average 0.6 customers when it arrives at device j .

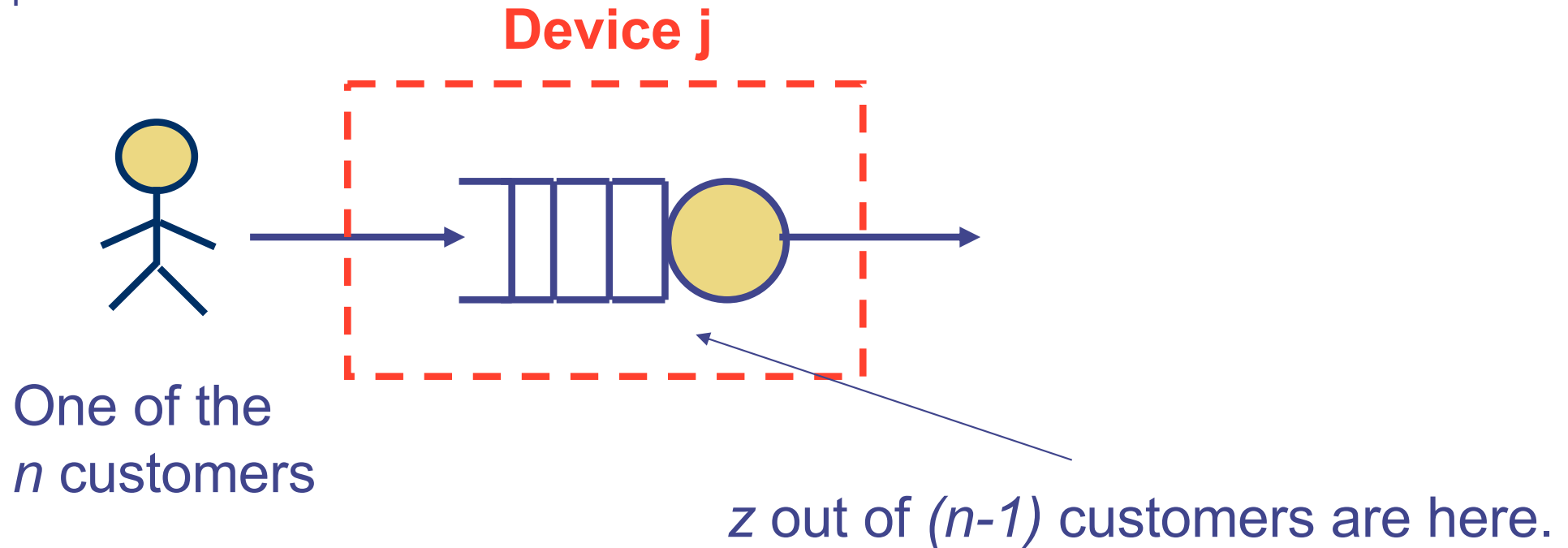
When there are n customers ...



Arrival Theorem

- If there are $(n-1)$ customers in the system, the mean number of customers in device j is z customers,
- Then, when there are n customers, each customer arriving at device j will see on average z customers ahead of itself in device j .

How can Arrival Theorem help?



Let $S(j)$ = mean service time at device j .

When there are n customers,

The mean waiting time at device $j = z S(j)$

The mean response time at device $j = (z+1) S(j)$

Iterations of MVA:

Mean number of customers in each device

#customers =
 $n-1$

Mean response time for each device

.....

.....

Mean number of customers in each device

#customers =
 n

Mean response time for each device

....

#customers =
 $n+1$

Some notation

Note " (n) " means there are n customers in the system

$\bar{n}_i(n)$ = Mean # of customers in device i

$R_i(n)$ = Mean response time in device i

$R_0(n)$ = Mean response time of the system

$X_i(n)$ = Throughput of device i

$X_0(n)$ = Throughput of the system

Mean response time of each device

$R_i(n)$

$$R_0(n) = \sum_{i=1}^K V_i \times R_i(n)$$

System response time

$R_0(n)$

$$X_0(n) = \frac{n}{R_0(n)}$$

Throughput of the system

$X_0(n)$

$$X_i(n) = V_i \times X_0(n)$$

Throughput of each device

$X_i(n)$

$$\bar{n}_i(n) = R_i(n) \times X_i(n)$$

Mean # customers in each device

$\bar{n}_i(n)$

Initialisation of MVA:

Mean number of customers in each device

#customers = 0

$$\bar{n}_i(0) = 0$$

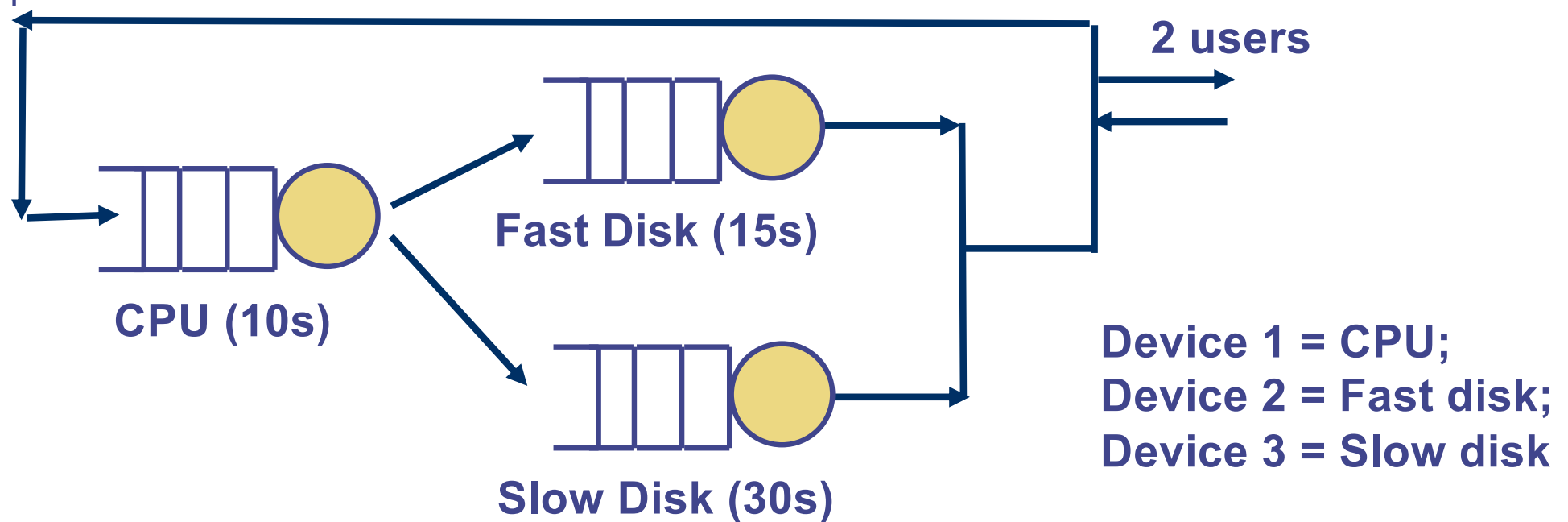
Mean response time for each device
.....
.....
Mean number of customers in each device

#customers = 1

Mean response time for each device
.....

#customers = 2

Let us apply MVA to the database server example



$$S_1 = 10; S_2 = 15; S_3 = 30;$$

$$V_1 = 1; V_2 = \frac{1}{2}; V_3 = \frac{1}{2};$$

- Determine the performance when there are 2 users in the system
- And how about 3 users?

Limitation of MVA

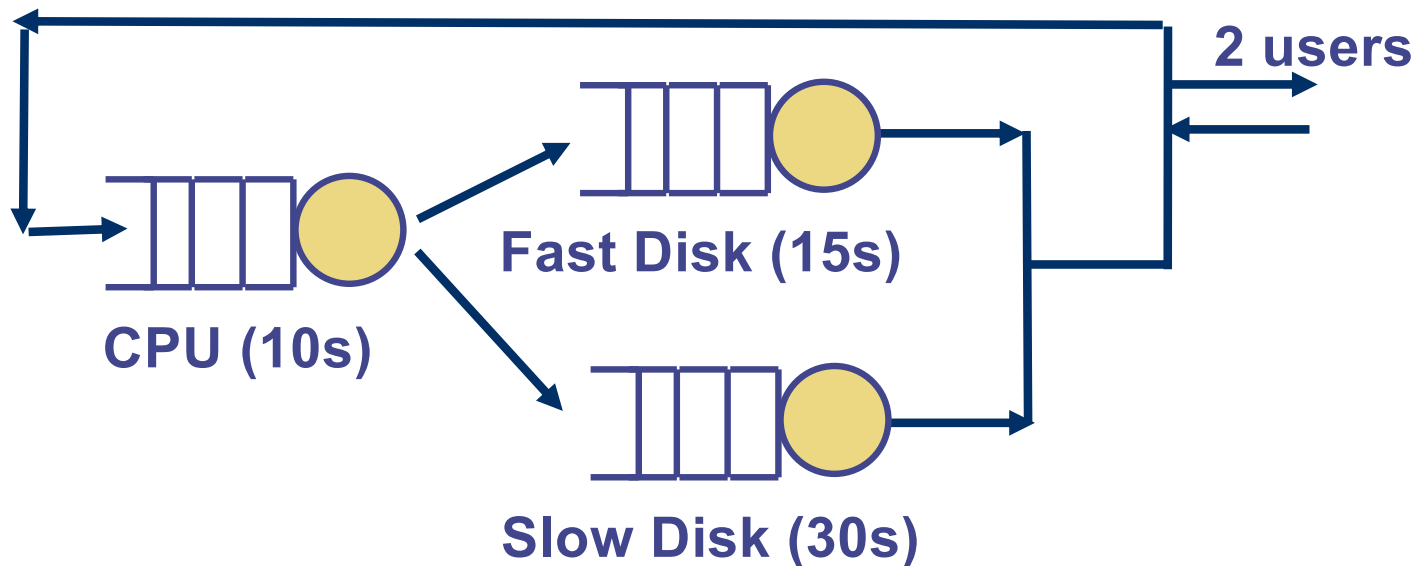
- MVA allows you to find the mean value of throughput, response time etc.
- However, if you are interested to find the probability that the system is in a certain state. MVA cannot give you the answer. You will need to resort to Markov model.

Extensions of MVA

- Closed queueing networks with multiple classes of customers
 - Example: Database servers with 2 classes of customers
 - One class of customers require mean service time of 0.02s, 0.03s and 0.05s from the CPU, fast and slow disk
 - Another class of customers require mean service time of 0.04s, 0.01s and 0.1s from the CPU, fast and slow disk
- Open queueing networks
- Mixed queueing networks

Assumptions behind MVA

- The service time is exponentially distributed
- The service time required at each component is independent
 - For example, MVA assumes that the service time required at CPU is independent of the service time at the disk

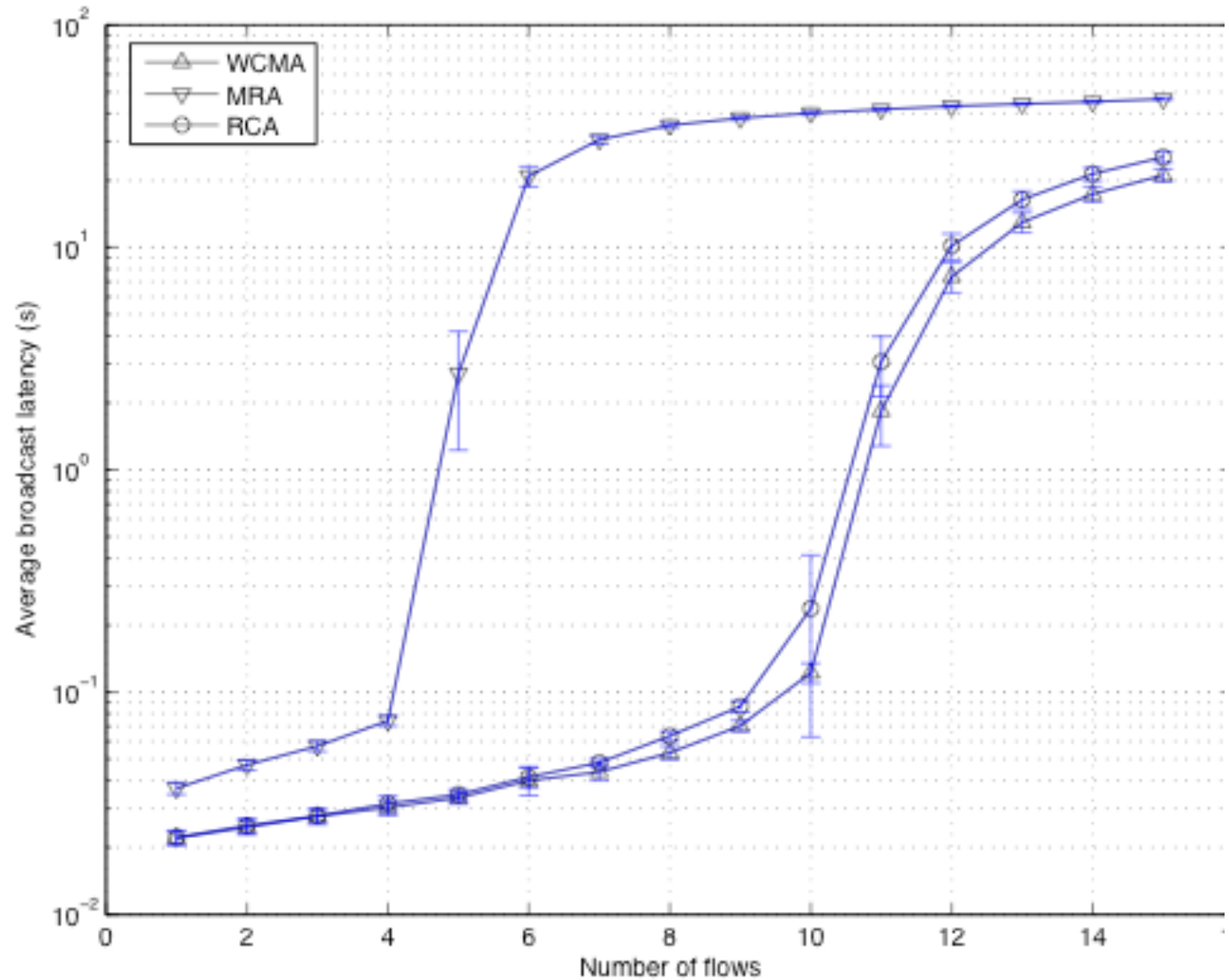


Solution to network of queues

- You have seen two possible methods to solve a network of queues
 - Analytical solution
 - Simulation
- For closed queueing networks with exponentially distributed service time
 - Markov chain
 - MVA
- Commercial simulation tools can deal with hundred of nodes

Multicast in wireless mesh networks

- In my research on designing multicast protocol for wireless mesh networks, we use simulation package *Qualnet* to investigate which of the multicast protocols that we have designed is better
- The network has 400 wireless mesh routers (= 400 queues)



- You can find out more on my research from my web site:
<http://www.cse.unsw.edu.au/~ctchou/>

Analytical solution versus simulation

- Analytical solution
 - Limited to specific cases
 - E.g. Exponential assumptions
 - Efficient computation algorithm exists for certain cases
 - MVA for closed queueing networks with exponential service time
- Simulation
 - Can apply to general settings
 - Difference classes of traffic, protocols etc.
 - Can apply to reasonably large networks too

References

- The primary reference for MVA for closed queueing networks with one class of customer is:
 - Chapter 12, Menasce et al., “Performance by design”
- An alternative reference for MVA is Chapter 6 of Edward Lazowska et al, Quantitative System Performance, Prentice Hall, 1984. (Now out of print but can be download from <http://www.cs.washington.edu/homes/lazowska/qsp/>)
 - Note that Chapter 6 has a wider coverage. It talks about open queueing network as well as approximation method too.
- For a formal mathematical proof of Arrival Theorem, see Bertsekas and Gallager, “Data networks”, Section 3.8.3